PRO-MODULARITY OF RESIDUALLY REDUCIBLE DEFORMATION RINGS OVER ABELIAN TOTALLY REAL FIELDS

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Allowed standard prerequisites (editorial).

- Original arithmetic geometry and number theory prerequisites. Original complete bibliography and source-local definitions retained.

1. Introduction

Let $\bar{\rho}_0 : \mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}) \to \mathrm{GL}_2(\mathbb{F})$ be a continuous odd representation, where $\mathbb{F}$ is finite field with characteristic $p \geq 3$. Let $R_{\bar{\rho}_0}$ be the universal deformation ring of $\bar{\rho}_0$. It is conjectured by Gouvêa ([Gou06]) that $R_{\bar{\rho}_0}$ is isomorphic to some p-adic big Hecke algebra. When $\bar{\rho}_0$ is absolutely irreducible, this question is studied by Böckle ([Böc01]) and Allen ([All19]). Recently, Deo has explored the reducible case in [Deo23]. In this article, we try to generalize his results to some totally real fields.

More precisely, suppose that $\bar{\rho}_0^{\mathrm{ss}} = 1 \oplus \bar{\chi}$ for some character $\bar{\chi}$ unramified outside Np, where N is a positive integer. Let $R_{\bar{\rho}_x}$ be the universal deformation ring of ρ_x, where $0 \neq x \in H^1(G_{\mathbb{Q},Np}, \mathbb{F}(\bar{\chi}))$ and ρ_x is the unique reducible Galois representation determined by x such that $\rho_x^{\mathrm{ss}} = 1 \oplus \bar{\chi}$. Let $R_{\bar{\rho}_0}^{\mathrm{pd}}$ be the universal pseudo-deformation ring of $\bar{\rho}_0^{\mathrm{ss}}$. Deo proves the following results.

Theorem 1.0.1. [Deo23, Theorem A, Theorem 4.5] *Let N_0 be the tame Artin conductor of $\bar{\rho}_0$. Suppose $N_0 | N$, $\bar{\chi}|_{G_{\mathbb{Q}_p}} \neq \mathbf{1}, \omega_p^{-1}$, $\dim_{\mathbb{F}} H^1(G_{\mathbb{Q},Np}, \mathbb{F}(\bar{\chi})) = 1$ and $p \nmid \phi(N)$. Then*

1) There is an isomorphism $(R_{\bar{\rho}_0}^{\mathrm{pd}})^{\mathrm{red}} \cong T(N)_{\bar{\rho}_0}$, where $T(N)_{\bar{\rho}_0}$ is the localization of the big Hecke algebra $T(N)$ (of level N) at the maximal ideal $\mathfrak{m}_0$ determined by $\bar{\rho}_0$.

2) There exists an isomorphism $R_{\bar{\rho}_x} \twoheadrightarrow T(N)_{\bar{\rho}_0}$.

In his proof, a crucial assumption is the cyclicity of $H^1(G_{\mathbb{Q},Np}, \mathbb{F}(\bar{\chi}))$. Under this assumption, Deo uses the fact that the natural map $R_{\bar{\rho}_0}^{\mathrm{pd}} \to R_{\bar{\rho}_x}$ is actually a surjection proved by Kisin [Kis09, Corollary 1.4.4] in the proof of 1), and constructs a surjection $R_{\bar{\rho}_x} \twoheadrightarrow T(N)_{\bar{\rho}_0}$ induced by $R_{\bar{\rho}_0}^{\mathrm{pd}} \twoheadrightarrow T(N)_{\bar{\rho}_0}$ by the theory of Generalized Matrix Algebra in the proof of

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2). However, such a cyclicity assumption is absurd when we consider a totally real field F instead of $\mathbb{Q}$ due to the global characteristic formula.

To generalize Deo's results to some totally real fields, we need a new strategy, which is inspired by the proof of the Fontaine-Mazur conjecture in the residually reducible case from Skinner-Wiles ([SW99]) and Pan ([Pan22]). Specifically, we study the pro-modularity of the universal deformation rings.

Now we state our main results.

Let $p > 3$ be a rational prime. Now we assume that F is an abelian totally real field of even degree such that p splits completely in F. Let Σ be a finite set of finite places of F containing Σ_p consisting of all the places $v \mid p$. Let $\mathcal{O}$ be the ring of integers of some local field E with a uniformizer π and residue field $\mathbb{F}$, and $\chi : \mathrm{Gal}(\bar{F}/F) \to \mathcal{O}^\times$ a totally odd character unramified outside Σ, and $\bar{\chi} = \chi \mod \pi$. Let F_Σ be the maximal extension of F unramified outside Σ and all infinite places. Let ϵ, ω_p be the p-adic, mod p cyclotomic character, respectively.

For an element $0 \neq x \in H^1(F_\Sigma/F, \mathbb{F}(\bar{\chi}^{-1}))$, let ρ_x be the unique non-split reducible Galois representation determined by x such that $\rho_x^{\mathrm{ss}} = 1 \oplus \bar{\chi}$. Let R_x^{aux} be the universal deformation ring of ρ_x of auxiliary type (see Definition 3.1.1). Let $R_{\mathrm{aux}}^{\mathrm{ps}}$ be the universal pseudo-deformation ring of the residual pseudo-representation $1+\bar{\chi}$ with fixed determinant χ. Let $\mathbb{T}_\xi$ be the big p-adic Hecke algebra (see Definition 4.1.1), and assume that $\bar{\rho}_0^{\mathrm{ss}}$ determines a maximal ideal $\mathfrak{m}_\xi$. For a prime $\mathfrak{p}$ of $R_{\mathrm{aux}}^{\mathrm{ps}}$ (resp. R_x^{aux}), we say that it is pro-modular if it comes from a prime of $(\mathbb{T}_\xi)_{\mathfrak{m}_\xi}$ (see Definition 4.2.3 and Definition 4.4.1 for the explicit definitions).

Theorem 1.0.2 (Theorem 4.4.2, Theorem 4.4.6 & Theorem 4.5.2)**.** *Keep notations as above. For any place $v \in \Sigma \setminus \Sigma_p$, assume that $p \mid \mathrm{Nm}(v) - 1$. Assume that the character χ satisfies the following conditions:*

1) χ is unramified outside Σ_p,

2) $\chi(\mathrm{Frob}_v) \equiv 1 \mod \pi$ for $v \in \Sigma \setminus \Sigma_p$.

3) $\bar{\chi}$ can be extended to a character of $G_{\mathbb{Q}}$.

4) $\bar{\chi} \mid_{G_{F_v}} \neq 1, \omega_p^{-1}$ for any $v \mid p$.

5) $\chi \mid_{G_{F_v}}$ is de Rham for any $v \mid p$.

(1) If $[F : \mathbb{Q}] \geq 7|\Sigma \setminus \Sigma_p| + 4$, then for any irreducible component of $R_{\mathrm{aux}}^{\mathrm{ps}}$ of dimension at least $1 + 2[F : \mathbb{Q}]$, it is pro-modular of dimension $1 + 2[F : \mathbb{Q}]$.

(2) If $[F : \mathbb{Q}] \geq \max\{1 + \frac{1}{2}\dim_{\mathbb{F}}(H^1(F_\Sigma/F, \mathbb{F}(\bar{\chi}^{-1}))), 7|\Sigma \setminus \Sigma_p| + 4\}$. Then for any non-zero cocycle $x \in H^1(F_\Sigma/F, \mathbb{F}(\bar{\chi}^{-1}))$, any prime ideal of R_x^{aux} is pro-modular. Further, R_x^{aux} is a local complete intersection ring of Krull dimension $1 + 2[F : \mathbb{Q}]$.

To prove such a theorem, we follow Pan's strategy in [Pan22], and a key tool is the patching argument (Theorem 4.2.5) at a nice prime (see Definition 4.2.3). Using Theorem 4.2.5, we only need to find a nice prime in each irreducible component of large dimension in the universal pseudo-deformation ring. It can be found successfully if the degree $[F : \mathbb{Q}]$ is large enough using Corollary 2.1.7.

Remark 1.0.3. One may ask how we can use these pro-modularity results.

The first part of Theorem 1.0.2, i.e., the pro-modularity of the universal pseudo-deformation ring, is actually a complement of the first step in Pan's proof of the Fontaine-Mazur conjecture in the residually reducible case (see [Pan22, page 1032]) so that after an abelian base change, the conjecture is a direct consequence of (1) of Theorem 1.0.2 and [Pan22, Corollary 3.5.10], although the full pro-modularity is not needed in his proof. In fact, Pan's strategy is to find a "potentially nice prime", i.e., it becomes nice after some soluble base change, and it is much easier to construct. Then he can identify some irreducible components of the universal (pseudo-)deformation ring as in the big Hecke algebra, which is enough for his purpose.

However, in the missing case of the Fontaine-Mazur conjecture in [Pan22], i.e., $p = 3$ and $\bar{\chi}|_{G_{F_v}} = \omega_p^{-1}$ for any $v \mid p$, to study the pro-modularity of some irreducible component of the universal (pseudo-)deformation ring, we need to know the pro-modularity of some other ones, which means we actually need a (potentially) big $R = \mathbb{T}$ theorem in this case. More precisely, the "dihedral condition" (the third one in (2) of Definition 4.2.3) sometimes obstructs the existence of the potentially nice primes, whereas the strategy of finding nice primes in this article, especially the application of Corollary 2.1.7, still works so that we just need to find pro-modular primes rather than nice primes (see Proposition 4.5.1). In other words, our method in this paper, combining some classical patching arguments, can be able to deal with some subtle cases of the conjecture, especially the case $p = 3$ and $\bar{\chi} = \omega_3$. Therefore, it allows us to drop Pan's second assumption of [Pan22, Theorem 1.0.4] and hence concludes the two-dimensional Fontaine-Mazur conjecture in the regular case for odd primes. See Remark 4.5.3 and [Zha24, Section 5] for more details.

We expect that our method can be generalized to prove more modularity or automorphy lifting theorems in the residually reducible case.

Remark 1.0.4. One may also ask whether we can find some examples such that our theorems may apply.

Our assumptions on the global character χ roughly state that it "comes from" the determinant of a continuous two-dimensional geometric Galois representation ρ of $G_{\mathbb{Q}}$. In

other words, starting from such a character $\chi = \det \rho$, using Grunwald-Wang's theorem (see [AT68, Theorem 5, page 80]), we can find a totally real abelian number field F satisfying all of our assumptions of (1) of Theorem 1.0.2. In other words, our pro-modularity result can be viewed as a potential big $R = \mathbb{T}$ theorem for general abelian totally real fields and global characters. This is also the way of applying this theorem to prove the Fontaine-Mazur conjecture.

Remark 1.0.5. We point out that compared to Deo's results, our pro-modularity result on the universal pseudo-deformation ring does not need the restrictive assumption about the dimension of the cohomology group.

Our pro-modularity result on the universal deformation ring is an analogue to Deo's result [Deo23, Theorem 4.5]. It is related to the dimension of the cohomology group since we need to bound the reducible locus of the deformation ring as in [SW99].

1.1. **Notations and conventions.** Throughout the paper, F is a totally real field and $\mathbb{A}_F$ is the adele ring. For a finite place v of F, we write F_v for the completion of F at v, $\mathcal{O}_{F_v}$ for its ring of integers, π_v for a uniformizer and $k(v)$ for the residue field. We denote the absolute Galois group and the inertia group of F_v by G_{F_v} and I_v, respectively. For a rational prime l, we denote $\mathcal{O}_F \otimes_{\mathbb{Z}} \mathbb{Z}_l$ by $\mathcal{O}_{F,l}$. We write $\mathrm{Nm}(v)$ for the norm of v, and write Frob_v for the geometric Frobenius element.

We fix a finite field $\mathbb{F}$ of characteristic $p > 3$. Let $\mathcal{O}$ be a ring of integers of some local field E (a finite extension of $\mathbb{Q}_p$) with a uniformizer π and residue field $\mathbb{F}$. Let $\mathscr{C}_{\mathcal{O}}^f$ be the category of Artinian local $\mathcal{O}$-algebras with residue field $\mathbb{F}$, and $\mathscr{C}_{\mathcal{O}}$ be the category of topological local $\mathcal{O}$-algebras which are isomorphic to inverse limits of objects of $\mathscr{C}_{\mathcal{O}}^f$. For a universal (pseudo-)deformation ring, we mean a universal object in $\mathscr{C}_{\mathcal{O}}$ which pro-represents the functor from $\mathscr{C}_{\mathcal{O}}^f$ to the category of sets sending R to the set of (pseudo-)deformations over R.

For a CNL ring R, we mean that R is a complete Noetherian local ring with maximal ideal $\mathfrak{m}_R$. If $\mathfrak{p}$ is a prime ideal of R, we will denote its residue field by $k(\mathfrak{p})$. We write $R_{\mathfrak{p}}$ for the localization at $\mathfrak{p}$, and $\widehat{R_{\mathfrak{p}}}$ for the $\mathfrak{p}$-adic completion. We write R^{red} for the maximal reduced quotient of R. By $M_i(R)$, we mean the ring of $i \times i$ matrices over R.

We write ϵ, ω_p for the p-adic, mod p cyclotomic character, respectively.

Acknowledgement. The author would like to express appreciation to his supervisor Professor Takeshi Saito for his careful reading, pointing out some mathematical errors and many helpful suggestions and comments.

2. Pseudo-representations

Pseudo-representations are the main tools to study the modularity of Galois representations in the residually reducible case. We recommend the article [BC09] to readers for a general theory of pseudo-representations.

2.1. **Two-dimensional pseudo-representations.** In this section, we will give some results on 2-dimensional pseudo-representations. Our main references are [SW99] and [Pan22].

Definition 2.1.1. For a profinite group G and a topological commutative ring R in which 2 is invertible, a 2-dimensional *pseudo-representation* is a continuous function $T : G \to R$ such that

1) $T(1) = 2$,

2) there exists an order 2 element $c \in G$ such that $T(c) = 0$,

3) $T(\sigma\tau) = T(\tau\sigma)$ for all $\sigma, \tau \in G$,

4) $T(\gamma\delta\eta) + T(\gamma\eta\delta) - T(\gamma\eta)T(\delta) - T(\eta\delta)T(\gamma) - T(\delta\gamma)T(\eta) + T(\gamma)T(\delta)T(\eta) = 0$, for any $\delta, \gamma, \eta \in G$.

The *determinant* $\det(T)$ of a pseudo-representation T is a character (using 4)) defined by

$$\det(T) : G \to R^\times,\ \det(T)(\sigma) = \frac{1}{2}(T(\sigma)^2 - T(\sigma^2)), \sigma \in G.$$

For a pseudo-representation T and $\sigma, \tau \in G$, define:

1) $a(\sigma) = \frac{1}{2}(T(c\sigma) + T(\sigma))$,

2) $d(\sigma) = \frac{1}{2}(-T(c\sigma) + T(\sigma))$,

3) $y(\sigma, \tau) = a(\sigma\tau) - a(\sigma)a(\tau)$.

If $\rho : G \to \mathrm{GL}_2(R)$ is a continuous representation with an element $c \in G$ satisfying $\rho(c) = \begin{pmatrix} 1 & \\ & -1 \end{pmatrix}$, then $\mathrm{tr}(\rho)$ is a pseudo-representation. More explicitly, if $\rho(\sigma) = \begin{pmatrix} a_\sigma & b_\sigma \\ c_\sigma & d_\sigma \end{pmatrix}$, then $a(\sigma) = a_\sigma$, $d(\sigma) = d_\sigma$ are as defined above and $y(\sigma, \tau) = b_\sigma c_\tau$.

Pan [Pan22, 2.1.4] verifies that the continuous functions $\{a, d, y\}$ satisfy the following equations.

(1) $y(\sigma, \tau) = d(\tau\sigma) - d(\tau)d(\sigma)$.

(2) $y(\sigma\tau, \delta) = a(\sigma)y(\tau, \delta) + y(\sigma, \delta)d(\tau)$.

(3) $y(\sigma, \tau\delta) = a(\delta)y(\sigma, \tau) + y(\sigma, \delta)d(\tau)$.

(4) $y(\alpha, \beta)y(\sigma, \tau) = y(\alpha, \tau)y(\sigma, \beta)$.

From the last equation, we can conclude the following two easy facts.

Proposition 2.1.2. Let R be an integral domain, and $T : G \to R$ be a pseudo-representation for a profinite group G. Let α, β be two elements in G.

1) If $y(\alpha, \beta) = 0$ in R, then either $y(\alpha, \tau) = 0$ or $y(\tau, \beta) = 0$ for any $\tau \in G$.

2) Assume that the set $J_\beta = \{\tau \in G : y(\beta, \tau) \neq 0\}$ (resp. $J'_\beta = \{\tau \in G : y(\tau, \beta) \neq 0\}$) is non-empty. For $\tau \in J_\beta$ (resp. J'_β), let $u(\alpha, \beta)(\tau) = \frac{y(\alpha,\tau)}{y(\beta,\tau)}$ (resp. $u'(\alpha, \beta)(\tau) = \frac{y(\tau,\alpha)}{y(\tau,\beta)}$) be an element in the fraction field of R. Then we have $u(\alpha, \beta)(\tau) = u(\alpha, \beta)(\tau')$ (resp. $u'(\alpha, \beta)(\tau) = u'(\alpha, \beta)(\tau')$) for any $\tau, \tau' \in J_\beta$ (resp. J'_β). (Afterwards, we will use notations $u(\alpha, \beta)$ and $u'(\alpha, \beta)$ for simplicity.)

Definition 2.1.3. [Pan22, 2.1.5] Assume that R is either a field or a discrete valuation ring. For a pseudo-representation $T : G \to R$, a representation ρ associated to T is in the following form.

1) $\rho(\sigma) = \begin{pmatrix} a(\sigma) & \\ & d(\sigma) \end{pmatrix}$, if all $y(\sigma, \tau) = 0$. We call this case reducible.

2) Choose σ_0, τ_0 such that $\frac{y(\sigma,\tau)}{y(\sigma_0,\tau_0)} \in R$ for any σ, τ, if $y(\sigma, \tau) \neq 0$ for some σ, τ. Define $\rho(\sigma) = \begin{pmatrix} a(\sigma) & \frac{y(\sigma,\tau_0)}{y(\sigma_0,\tau_0)} \\ y(\sigma_0, \sigma) & d(\sigma) \end{pmatrix}$. We call this case irreducible.

Remark 2.1.4. We can check that ρ is actually absolutely irreducible, and is unique up to conjugation if R is a field in the irreducible case.

The following results are essential to our pro-modularity arguments.

Proposition 2.1.5. Let R be a CNL domain with maximal ideal $\mathfrak{m}_R$ and residue field $\mathbb{F}$. Let G be a profinite group and $T : G \to R$ be a pseudo-representation of G. Assume that $T \mod \mathfrak{m}_R$ is reducible. Let α be an element of G and write I_α for the ideal of R generated by $\{y(\alpha, \beta), \beta \in G\}$. Assume that any minimal prime belonging to I_α (following [AM69, page 52]) is of height at least 2.

1) For any $\sigma \in G$, $u(\sigma, \alpha)$ (defined in Proposition 2.1.2) is well-defined and integral over R.

2) If α' is an another element of G such that any minimal prime belonging to $I_{\alpha'}$ of R generated by $\{y(\alpha', \beta), \beta \in G\}$ is of height at least 2, then $u(\alpha, \alpha')$ is a unit in the normal closure of R.

Proof. Let B be the normal closure of R, and $i : R \hookrightarrow B$ be the natural inclusion. As R is a CNL domain, by [Sta23, Lemma 15.106.3], R is unibranch. Thus, B is also a local domain.

We first recall a useful result in commutative algebra.

Lemma 2.1.6. Let $\mathfrak{p}$ be a prime ideal of B of height d. Then $i^{-1}(\mathfrak{p})$ is also of height d in R.

Proof. This is [Eis13, Proposition 9.2].

□

Now we return to the proof of our proposition. As $T \mod \mathfrak{m}_R$ is reducible, we know that for all $\alpha, \beta \in G$, we have $y(\alpha, \beta) \in \mathfrak{m}_R$.

1) By Lemma 2.1.6, we know that there does not exist a prime of B containing $\{y(\alpha, \beta), \beta \in \mathrm{Gal}(F_\Sigma/F)\}$ of height at most 1. As B is normal , we have $\cap_{\mathfrak{p}_B} B_{\mathfrak{p}_B} = B$, where $\mathfrak{p}_B$ ranges over all height one primes of B. To prove our result, for any height one prime $\mathfrak{p}_B$, we just need to find an elemnt $\beta \in G$ satisfying $\frac{y(\sigma,\beta)}{y(\alpha,\beta)} \in B_{\mathfrak{p}_B}$, and it is immediate from our discussion. This proves 1).

2) By 1), we know that both $u(\alpha', \alpha)$ and $u(\alpha, \alpha')$ are well-defined and integral over R. As $u(\alpha', \alpha)u(\alpha, \alpha') = 1$, we know that $u(\alpha, \alpha')$ is a unit in B. □

Corollary 2.1.7. *Keep notations as in the statement of Proposition 2.1.5. Let S be a finite subset (not necessarily a group) of G . Then there exist a partition of $S = S_1 \amalg S_2$, a positive integer $n > 1$ satisfying $(n, p) = 1$ and a CNL domain R' satisfying the following conditions.*

a) R' is a quotient of R.

b) For any $\theta \in S_1$, we have $y(\theta, \alpha) = 0$ for any $\alpha \in G$ in R'. For any $\theta', \theta'' \in S_2$, we have $y(\theta', \alpha)^n = y(\theta'', \alpha)^n$ for any $\alpha \in G$ in R'.

c) For any $\alpha \in G$, $\theta \in S_2$, either $y(\theta, \beta) = 0$ for any $\beta \in G$ in R' or $u(\alpha, \theta)$ is well-defined and integral over R'.

d) We have $\dim R' \geq \dim R - |S|$. *If S_2 is not empty, then further* $\dim R' \geq \dim R - |S| + 1$.

Proof. We prove this result by induction on $|S|$.

If $|S| = \varnothing$, then our result is just trivial.

Now we assume that our result holds for a non-negative integer k and $|S| = k + 1$.

First, we check whether there exists an element $\theta \in S$ such that there exists a minimal prime belonging to the ideal I_θ of R generated by $\{y(\theta, \beta), \beta \in G\}$ is of height at most 1. Recall that B is the normal closure of R.

If not, we can take $S_2 = S$ and $n = |B/\mathfrak{m}_B| - 1$ (clearly prime to p). Now we fix an element $\theta_1 \in S_2$. For any $\theta' \in S_2$, by Proposition 2.1.5, $u(\theta_1, \theta')$ is well-defined and integral over R, and hence a unit in B. Furthermore, $u(\theta_1, \theta')^n - 1$ is an element of the maximal ideal of B. Consider the ideal $\mathfrak{b}$ of B generated by $\{u(\theta_1, \theta')^n - 1, \theta' \in S_2\}$. By Krull's principal ideal theorem, we can find a prime $\mathfrak{p}$ belonging to $\mathfrak{b}$ of height at most $|S_2| - 1$. By Lemma 2.1.6, its inverse image $\mathfrak{p}'$ in R is also of height at most $|S_2| - 1$. Then we can take $R' = R/\mathfrak{p}'$. The four conditions follow from our construction, Proposition 2.1.2 and Proposition 2.1.5.

If so, we can choose such an element $\theta \in S$ and consider a prime $\mathfrak{p}_\theta$ (of R) of I_θ of height at most 1. Take $R_1 = R/\mathfrak{p}_\theta$. Then we have $y(\theta, \alpha) = 0$ for any $\alpha \in G$ and $\dim R_1 \geq \dim R - 1$. Now we can use our induction hypothesis for the CNL domain R_1 and the finite set $S \setminus \{\theta\}$. Then we get our desired result. □

Remark 2.1.8. This result is also used in [Zha24]. See [Zha24, Lemma 4.2.5 & 5.1.2].

3. Galois deformation theory

In this section, we will study some properties of 2-dimensional universal deformation rings and pseudo-deformation rings.

3.1. Universal deformation rings.

In this subsection, we study some properties of universal deformation rings related to ρ_x.

Let F be a totally real field. Let $p > 3$ be a rational prime and $\mathbb{F}$ be a finite field of characteristic p. Let Σ be a finite set of finite places of F containing all the places $v \mid p$. Recall that $\mathcal{O}$ is the ring of integers of some local field E (a finite extension of $\mathbb{Q}_p$) with a uniformizer π and residue field $\mathbb{F}$. Let $\chi : \mathrm{Gal}(\bar{F}/F) \to \mathcal{O}^\times$ be a continuous totally odd character unramified outside Σ, and $\bar{\chi} = \chi \mod \pi$. Let F_Σ be the maximal extension of F unramified outside Σ and all infinite places. We denote $\bar{\rho}_0$ by the representation $1 \oplus \bar{\chi}$.

For an element $0 \neq x \in H^1(F_\Sigma/F, \mathbb{F}(\bar{\chi}^{-1}))$, there is a unique representation

$$\rho_x : \mathrm{Gal}(F_\Sigma/F) \to \mathrm{GL}_2(\mathbb{F}),\ \rho_x = \begin{pmatrix} 1 & * \\ & \bar{\chi} \end{pmatrix}$$

such that $\rho_x(c) = \begin{pmatrix} 1 & \\ & -1 \end{pmatrix}$ for a complex conjugation c, where $*$ corresponds to x.

Let R_x be the universal deformation ring in the sense of Mazur [Maz89] of ρ_x and $\rho_x^{\mathrm{univ}} : \mathrm{Gal}(F_\Sigma/F) \to \mathrm{GL}_2(R_x)$ the universal deformation of ρ_x. Following [SW99, Proposition 2.4], we consider the deformation problem of auxiliary type.

Definition 3.1.1. A deformation $\rho : \mathrm{Gal}(\bar{F}/F) \to \mathrm{GL}_2(A)$ of ρ_x is of auxiliary type if

1) A is an $\mathcal{O}$-algebra,
2) $\det \rho = \chi$,
3) ρ factors through $\mathrm{Gal}(F_\Sigma/F)$.

We denote the universal deformation of auxiliary type by $\rho_x^{\mathrm{aux}} : \mathrm{Gal}(F_\Sigma/F) \to \mathrm{GL}_2(R_x^{\mathrm{aux}})$.

Proposition 3.1.2. The ring R_x^{aux} is isomorphic to $\mathcal{O}[[x_1, \cdots, x_g]]/(f_1, \cdots, f_r)$, where $g = \dim_{\mathbb{F}}(H^1(F_\Sigma/F, \mathrm{ad}^0 \rho_x))$, $r \leq \dim_{\mathbb{F}}(H^2(F_\Sigma/F, \mathrm{ad}^0 \rho_x))$.

Proof. The assertion is from [Maz89, Proposition 2]. □

Proposition 3.1.3. The dimension of every irreducible component of R_x^{aux} is at least $2[F : \mathbb{Q}] + 1$.

Proof. This is a direct consequence of Proposition 3.1.2, Krull's principal ideal theorem and the global characteristic formula. □

Let $\rho_x^{\mathrm{aux,red}} : \mathrm{Gal}(F_\Sigma/F) \to \mathrm{GL}_2(R_x^{\mathrm{aux,red}})$ be the universal reducible deformation of ρ_x of auxiliary type. Clearly, $R_x^{\mathrm{aux,red}}$ is a quotient of R_x^{aux}.

Proposition 3.1.4. Let δ_F be the Leopoldt defect (see [NSW13, 10.3.7]. Then we have $\dim(R_x^{\mathrm{aux,red}}) \leq 1 + \delta_F + \dim_{\mathbb{F}}(H^1(F_\Sigma/F, \mathbb{F}(\bar{\chi}^{-1})))$.

Proof. The proof is an analogue to [SW99, Section 2.2]. We sketch a proof here.

Let $L(\Sigma)$ is the maximal abelian pro-p-extension of F unramified outside Σ, and write $G = \mathrm{Gal}(L(\Sigma)(\bar{\chi})/F) \simeq \Delta \times \Gamma \times \mathbb{Z}_p^{\delta_F+1}$, where $\Delta = \mathrm{Gal}(F(\bar{\chi})/F))$, Γ is a finite p-group, $L(\Sigma)(\bar{\chi}) = L(\Sigma) \cdot F(\bar{\chi})$, and $F(\bar{\chi})$ is the splitting field. Let M be the maximal abelian pro-p-extension of $L(\Sigma)(\bar{\chi})$ unramified outside Σ such that Δ acts on $H = \mathrm{Gal}(M/L(\Sigma)(\bar{\chi}))$ via $\bar{\chi}^{-1}$. Then any reducible deformation of auxiliary type factors through $\mathrm{Gal}(M/F)$.

Put $A = \mathbb{Z}_p[[G]]$. By Nakayama's lemma, the group H is a finitely generated A-module generated by $m := \dim_{\mathbb{F}}(H^1(F_\Sigma/F, \mathbb{F}(\bar{\chi}^{-1})))$ elements. Fix a presentation $\mathfrak{a} \to \bigoplus_{i=1}^m Ae_i \twoheadrightarrow H$ such that e_m projects to $h_m \in H$ for which $\rho_x(h_m) = \begin{pmatrix} 1 & u \\ & 1 \end{pmatrix}$ with $u \neq 0$ and such that if $i \neq m$ then e_i projects to $h_i \in H$ for which $\rho_x(h_i) = \begin{pmatrix} 1 & \\ & 1 \end{pmatrix}$. Choose a $u_0 \in \mathcal{O}^\times$ reducing to u. Put $A_1 = \mathcal{O}[[\Gamma]][[T_1, \cdots, T_{\delta_F+1}]]$ and fix an embedding $\phi : A \hookrightarrow A \otimes_{\mathbb{Z}_p[\Delta]} \mathcal{O} \simeq A_1$ where the map $\mathbb{Z}_p[\Delta] \to \mathcal{O}$ is via $\bar{\chi}$. Let J be the ideal of $A_1[[x_1, \cdots, x_{m-1}]]$ generated by

$$\{\phi(a_1)x_1 + \cdots + \phi(a_{m-1})x_{m-1} + \phi(a_m)u_0 : \Sigma a_i e_i \in \mathfrak{a}\}.$$

Let $B = A_1[[x_1, \cdots, x_{m-1}]]/J$. Define $\rho : \mathrm{Gal}(M/F) \to \mathrm{GL}_2(B)$ by

$$\rho(g) = \begin{pmatrix} \chi(g) \cdot \phi(g)^{-1} & \\ & \phi(g) \end{pmatrix}, \ g \in G,$$

$$\rho(h) = \begin{pmatrix} 1 & \tau(h) \\ & 1 \end{pmatrix}, \tau(h_i) = e_i, \ i = 1, \cdots, m-1, \ \tau(h_m) = u_0.$$

By the universal property, we know that $B \cong R_x^{\mathrm{aux,red}}$. Hence, we have $\dim(R_x^{\mathrm{aux,red}}) \leq 1 + \delta_F + 1 + m - 1 = 1 + \delta_F + \dim_{\mathbb{F}}(H^1(F_\Sigma/F, \mathbb{F}(\bar{\chi}^{-1})))$.

□

Proposition 3.1.5. Assume that for all $v \in \Sigma \setminus \Sigma_p$, we have $p \mid \mathrm{Nm}(v) - 1$, and χ is unramified outside Σ_p. Let R be a CNL domain with maximal ideal $\mathfrak{m}_R$, fraction field K and residue field $\mathbb{F}$. Suppose that $\rho : \mathrm{Gal}(F_\Sigma/F) \to \mathrm{GL}_2(R)$ is a deformation of ρ_x of auxiliary type. Then for any $v \in \Sigma \setminus \Sigma_p$, there exists a finite extension K'/K and a finite character $\xi_v : I_v \to K'^\times$ of p-power order such that $\mathrm{tr}(\rho)\,|_{I_v} = \xi_v + \xi_v^{-1}$.

Proof. It is an analogue to [Pan22, Corollary 5.7.2]. We sketch a proof here.

As $v \nmid p$, we know that for any $\sigma \in I_v$, $\rho(\sigma)$ is of pro-p-order. From the structure of Galois group of the maximal tamely ramified extension [NSW13, Proposition 7.5.2], we know that $\rho\,|_{I_v}$ only depends on the $\mathbb{Z}_p$-quotient of I_v, and we denote the generator by σ_v.

As ρ is of auxiliary type, we just need to show that the eigenvalues α, α^{-1} of $\rho(\sigma_v)$ are of p-power order. By [NSW13, Theorem 7.5.3], there exists an element in $\tau \in G_{F_v} \setminus I_v$ satisfying $\tau\sigma_v\tau^{-1} = \sigma_v^{\mathrm{Nm}(v)}$. Thus, we have $\alpha^{(\mathrm{Nm}(v)^2-1)} = 1$. Since $\alpha \equiv 1 \mod \mathfrak{m}_R$, we know that α is of p-power order. □

Now we suppose that $\mathcal{O}$ has enough p-power roots of unity with residue field $\mathbb{F}$. Let $\xi_v : k(v)^\times \to \mathcal{O}^\times$ be characters of p-power order for $v \in \Sigma \setminus \Sigma_p$, and we can view them as characters of I_v by local class field theory.

Now consider the deformation ρ of ρ_x such that

1) it is of auxiliary type,

2) it is unramified outside Σ,

3) $\mathrm{tr}(\rho)\,|_{I_v} = \xi_v + \xi_v^{-1}$ for all $v \in \Sigma \setminus \Sigma_p$.

Let $\rho_x^{\mathrm{aux},\{\xi_v\}} : \mathrm{Gal}(F_\Sigma/F) \to \mathrm{GL}_2(R_x^{\mathrm{aux},\{\xi_v\}})$ be the universal deformation of such a deformation problem. Obviously, $R_x^{\mathrm{aux},\{\xi_v\}}$ is a quotient of R_x^{aux}.

By Proposition 3.1.5, we know that for any minimal prime $\mathfrak{P}$ of R_x^{aux}, the natural surjection $R_x^{\mathrm{aux}} \to R_x^{\mathrm{aux}}/\mathfrak{P}$ factors through some $R_x^{\mathrm{aux},\{\xi_v\}}$. We say that $\mathfrak{P}$ *belongs to* $\{\xi_v\}$ in this situation.

3.2. **Universal pseudo-deformation ring.** In this subsection, we collect some results which will be useful for our pro-modularity arguments.

Keep the notations and assumptions in the previous subsection. Write $\Sigma_p \subset \Sigma$ as the set of places of F lying above p. We further assume that for all $v \in \Sigma \setminus \Sigma_p$, we have $p \mid \mathrm{Nm}(v) - 1$, and that $\chi : \mathrm{Gal}(F_\Sigma/F) \to \mathcal{O}^\times$ such that

1) χ is unramified outside Σ_p,

2) $\chi(\mathrm{Frob}_v) \equiv 1 \mod \pi$ for $v \in \Sigma \setminus \Sigma_p$.

Let R^{ps} be the universal pseudo-deformation ring of $\mathrm{tr}(\bar{\rho}_0) = 1 + \bar{\chi}$, and T^{univ} the universal deformation. The existence of $(R^{\mathrm{ps}}, T^{\mathrm{univ}})$ is proved in [SW99, Section 2.4]. Let

$(R^{\mathrm{ps}}_{\mathrm{aux}}, T_{\mathrm{aux}})$ be the universal pseudo-deformation of $\mathrm{tr}(\bar{\rho}_0)$ of auxiliary type (i.e., with $\det = \chi$) and $(R^{\mathrm{ps},\{\xi_v\}}, T^{\mathrm{ps},\{\xi_v\}})$ the universal pseudo-deformation of $\mathrm{tr}(\bar{\rho}_0)$ of auxiliary type satisfying $T_{\mathrm{aux}}\mid_{I_v} = \xi_v + \xi_v^{-1}$ for all $v \in \Sigma \setminus \Sigma_p$. Clearly, there are natural surjections $R^{\mathrm{ps}} \twoheadrightarrow R^{\mathrm{ps}}_{\mathrm{aux}} \twoheadrightarrow R^{\mathrm{ps},\{\xi_v\}}$. By [Pan22, Corollary 5.7.2], for any minimal prime $\mathfrak{P}$ of $R^{\mathrm{ps}}_{\mathrm{aux}}$, the natural surjection $R^{\mathrm{ps}}_{\mathrm{aux}} \to R^{\mathrm{ps}}_{\mathrm{aux}}/\mathfrak{P}$ factors through some $R^{\mathrm{ps},\{\xi_v\}}$. Similarly, we also say that $\mathfrak{P}$ *belongs to* $\{\xi_v\}$ in this situation.

From our definitions, we can get the following results immediately.

Proposition 3.2.1. 1) $R_x^{\mathrm{aux},\{\xi_v\}}/\pi = R_x^{\mathrm{aux},\{\xi_v'\}}/\pi$.

2) $R^{\mathrm{ps},\{\xi_v\}}/\pi = R^{\mathrm{ps},\{\xi_v'\}}/\pi$.

Let $(R^{\mathrm{ps,red}}, T^{\mathrm{red}})$ be the universal pseudo-deformation ring of auxiliary type such that T^{red} is reducible, i.e., $y(\sigma,\tau) = 0$ for all $\sigma, \tau \in \mathrm{Gal}(F_\Sigma/F)$.

Proposition 3.2.2. Assume F is abelian over $\mathbb{Q}$.

1) We have $\dim(R^{\mathrm{ps,red}}) \leq 2$.

2) Assume $\bar{\chi}$ is quadratic and can be extended to $G_{\mathbb{Q}}$. Then the dihedral locus C^{dih} of $\operatorname{Spec} R^{\mathrm{ps}}_{\mathrm{aux}}$, i.e., for any point $\mathfrak{p}$ in C^{dih}, $\rho(\mathfrak{p}) \cong \mathrm{Ind}_{G_L}^{G_F}\theta$ for some character θ and the splitting field $L = F(\bar{\chi})$, is of dimension at most $2 + [F:\mathbb{Q}]$.

Proof. 1) Consider the character $d : \mathrm{Gal}(F_\Sigma/F) \to (R^{\mathrm{ps,red}})^\times$ in the sense of Definition 2.1.1, and it induces a natural surjection $\mathcal{O}[[G_F^{ab}]] \twoheadrightarrow R^{\mathrm{ps,red}}$. Hence, $\dim(R^{\mathrm{ps,red}}) \leq 2 + \delta_F = 2$. The last equation is from Leopoldt's conjecture for abelian totally real fields.

2) From our definition, we can find that if $\mathfrak{p} \in C^{\mathrm{dih}}$ and $\rho(\mathfrak{p}) \cong \mathrm{Ind}_{G_L}^{G_F}\theta$, then the global character θ is a lifting of the character $\bar{\chi}|_{G_L} = \mathbf{1}$. Thus, the dihedral locus is at most of $\dim \mathcal{O}[[G_L^{ab}]]$, which is $2 + [F:\mathbb{Q}]$ by Leopoldt's conjecture for abelian CM fields. □

Corollary 3.2.3. *Keep assumptions as in Proposition 3.2.2. Let $\mathfrak{s}$ be a prime of $R^{\mathrm{ps}}_{\mathrm{aux}}$ containing p and write $R = R^{\mathrm{ps}}_{\mathrm{aux}}/\mathfrak{s}$.*

1) If $\dim R \geq 2$, then we can find a prime $\mathfrak{s}'$ of $R^{\mathrm{ps}}_{\mathrm{aux}}$ containing $\mathfrak{s}$ such that the corresponding representation is irreducible.

2) If $\dim R \geq 2 + [F:\mathbb{Q}]$, then we can find a prime $\mathfrak{s}'$ of $R^{\mathrm{ps}}_{\mathrm{aux}}$ containing $\mathfrak{s}$ such that the corresponding representation is irreducible and not dihedral.

Proof. We first give a useful lemma in commutative algebra.

Lemma 3.2.4. *Let R be a noetherian local ring with maximal ideal $\mathfrak{m}_R$. Suppose that $V \subsetneqq \operatorname{Spec} R$ is a closed subset. Then there exists a one-dimensional prime $\mathfrak{p} \notin V$.*

Proof of Lemma 3.2.4. Using [Sta23, Lemma 28.6.4], the scheme $\operatorname{Spec} R \setminus \{\mathfrak{m}_R\}$ is Jacobson, i.e., the underlying topological space is Jacobson. As $V \neq \operatorname{Spec} R$, there exists a closed point of $\operatorname{Spec} R \setminus \{\mathfrak{m}_R\}$ not contained in V. This shows the result. □

Our assertions are direct consequences from Lemma 3.2.4 and Proposition 3.2.2. □

Following [SW99] and [Pan22], it is natural to establish the relationship between deformation rings and pseudo-deformation rings.

Proposition 3.2.5. 1) For any one-dimensional prime ideal $\mathfrak{p}_x \in \operatorname{Spec} R_x$ such that $\rho^{\mathrm{univ}} \bmod \mathfrak{p}_x$ is irreducible, the natural map $\widehat{(R^{\mathrm{ps}}_{\mathfrak{p}})} \to \widehat{(R_x)_{\mathfrak{p}_x}}$ is an isomorphism, and $\operatorname{Spec}(R_x)_{\mathfrak{p}_x} \to \operatorname{Spec} R^{\mathrm{ps}}_{\mathfrak{p}}$ is surjective. Here $\mathfrak{p} = \mathfrak{p}_x \cap R^{\mathrm{ps}}$.

2) For any $\mathcal{Q} \in \operatorname{Spec} R_x$ such that $\rho^{\mathrm{univ}} \bmod \mathcal{Q}$ is irreducible, we have

$$\dim R_x/\mathcal{Q} \leq \dim R^{\mathrm{ps}}/(\mathcal{Q} \cap R^{\mathrm{ps}}).$$

Moreover, if $\mathcal{Q}$ is a minimal prime, so is $\mathcal{Q} \cap R^{\mathrm{ps}}$.

3) The results in 1), 2) are also true if we replace (R_x, R^{ps}) by $(R^{\mathrm{aux}}_x, R^{\mathrm{ps}}_{\mathrm{aux}})$.

Proof. The first two assertions are [Pan22, Corollary 2.3.6]. The proof of the last assertion is the same as Pan's.

□

Definition 3.2.6. [Pan22, Definition 5.4.3] Let R be a CNL ring. The connectedness dimension of R is defined to be

$$c(R) := \min_{Z_1, Z_2} \{\dim(Z_1 \cap Z_2)\},$$

where Z_1, Z_2 are non-empty unions of irreducible components of $\operatorname{Spec} R$ such that $Z_1 \cup Z_2 = \operatorname{Spec} R$.

Proposition 3.2.7. Suppose $\mathfrak{p}$ is a maximal ideal of $R^{\mathrm{ps}}_{\mathrm{aux}}[1/p]$ whose corresponding representation is absolutely irreducible.

1) $\dim(R^{\mathrm{ps}}_{\mathrm{aux}})_{\mathfrak{p}} \geq 2[F : \mathbb{Q}]$.

2) The connectedness dimension $c((R^{\mathrm{ps}}_{\mathrm{aux}})_{\mathfrak{p}})$ is at least $2[F : \mathbb{Q}] - 1$.

Proof. For the first part, see [Pan22, Lemma 7.1.2]. For the second part, see [Pan22, Lemma 7.4.6]. □

Remark 3.2.8. Unlike the universal deformation ring R^{aux}_x, the structure of the universal pseudo-deformation ring $R^{\mathrm{ps}}_{\mathrm{aux}}$ is mysterious because we cannot use the global characteristic formula as in Proposition 3.1.2. However, we can show that if $\mathfrak{p}$ is a one-dimensional

irreducible prime of $R^{\mathrm{ps}}_{\mathrm{aux}}$, then every irreducible component of $\operatorname{Spec} R^{\mathrm{ps}}_{\mathrm{aux}}$ containing $\mathfrak{p}$ is of dimension at least $2[F:\mathbb{Q}]+1$, as an analogue to Proposition 3.1.3. The proof is as follows.

If our claim is not true, then we have a partition $Z_1 \amalg Z_2 = \operatorname{Spec}(R^{\mathrm{ps}}_{\mathrm{aux}})_{\mathfrak{p}}$, where Z_i are finite sets of some irreducible components of $\operatorname{Spec}(R^{\mathrm{ps}}_{\mathrm{aux}})_{\mathfrak{p}}$ such that Z_1 consists of all ones of dimension at least $2[F:\mathbb{Q}]$. By the first part of Proposition 3.2.7 and our assumption, both Z_1 and Z_2 are non-empty. By the second part of Proposition 3.2.7, we can find $C_1 \in Z_1, C_2 \in Z_2$ and a prime $\mathfrak{q} \in C_1 \cap C_2$ of dimension at least $2[F:\mathbb{Q}]-1$. As $\dim C_2 \leq 2[F:\mathbb{Q}]-1$, we know that $\mathfrak{q}$ is the generic point of C_2, which is against to the fact $\mathfrak{q} \in C_1$. Thus, Z_2 has to be empty.

We do not know whether every irreducible component of $R^{\mathrm{ps}}_{\mathrm{aux}}$ is of dimension at least $2[F:\mathbb{Q}]+1$.

4. Pro-modularity

In this section, we will prove the pro-modularity of the universal deformation ring. We fix a totally real field F and a prime $p>3$.

4.1. **Hecke algebras.** In this subsection, we introduce Hecke operators and Hecke algebras. The main reference is [Pan22, Section 3].

Assume that the prime p splits completely in the totally real field F.

Let D be a quaternion algebra with centre F which is ramified at all infinite places of F and unramified at all places above p, and we fix isomorphisms $D \otimes F_v \cong M_2(F_v)$ for any v where D is unramified. We view $K_p = \prod_{v|p} \mathrm{GL}_2(\mathcal{O}_{F_v})$ and $D_p^\times = \prod_{v|p} \mathrm{GL}_2(F_v)$ as subgroups of $(D \otimes_F \mathbb{A}_F)^\times$.

Let A be a topological $\mathcal{O}$-algebra and $U = \prod_{v|p} U_v$ be an open compact subgroup of $(D \otimes_F \mathbb{A}_F^\infty)^\times$ such that $U_v \subseteq \mathrm{GL}_2(\mathcal{O}_{F_v})$ for $v \mid p$. We write $U^p = \prod_{v\nmid p} U_v$ (tame level) and $U_p = \prod_{v|p} U_v$. Let $\psi : (\mathbb{A}_F^\infty)^\times / F_+^\times \to A^\times$ be a continuous character, where F_+ is the set of totally positive elements in F. Let $\tau : \prod_{v|p} U_v \to \mathrm{Aut}(W_\tau)$ be a continuous representation on a finite A-module W_τ, and we also view τ as a representation of U by projecting to $\prod_{v|p} U_v$.

Suppose that $\xi : U^p \to A^\times$ is a continuous smooth character such that $\psi \mid_{\prod_{v\nmid p}(\mathcal{O}_{F_v}^\times)\cap U_v} = \xi \mid_{\prod_{v\nmid p}(\mathcal{O}_{F_v}^\times)\cap U_v}$. We define $S_{\tau,\psi,\xi}(U,A)$ as the space of continuous functions: $f : D^\times \setminus (D \otimes_F \mathbb{A}_F^\infty)^\times \to W_\tau$, such that for any $g \in (D \otimes_F \mathbb{A}_F^\infty)^\times$, $z \in (\mathbb{A}_F^\infty)^\times$, $u = u^p u_p \in U$, we have

$$f(guz) = \psi(z)\xi(u^p)\tau(u_p^{-1})(f(g)).$$

If $\psi \mid_{U_p \cap \mathcal{O}_{F,p}^\times} = \tau^{-1} \mid_{U_p \cap \mathcal{O}_{F,p}^\times}$, then we have

$$S_{\tau,\psi,\xi}(U,A) \simeq \bigoplus_{i\in I} W_\tau^{(t_i^{-1} D^\times t_i \cap U(\mathbb{A}_F^\infty)^\times)/F^\times},$$

where $I = D^\times \setminus (D \otimes_F \mathbb{A}_F^\infty)^\times / U(\mathbb{A}_F^\infty)^\times$ and $\{t_i\}_{i\in I}$ is a set of representatives. We will simply write $S_{\psi,\xi}(U, A)$ if τ is the trivial action on A. We say that U is *sufficiently small* if $(t_i^{-1} D^\times t_i \cap U(\mathbb{A}_F^\infty)^\times)/F^\times$ is trivial for all $i \in I$.

Let Σ be a finite set of primes of F containing all places above p and places v where either D is ramified or U_v is not a maximal open subgroup. For any $v \notin \Sigma$, we define the Hecke operator $T_v \in \mathrm{End}(S_{\tau,\psi,\xi}(U, A))$ to be the double coset action $[U_v \begin{pmatrix} \pi_v & \\ & 1 \end{pmatrix} U_v]$. Precisely, if we write $U_v \begin{pmatrix} \pi_v & \\ & 1 \end{pmatrix} U_v = \coprod_i \gamma_i U_v$, then

$$(T_v \cdot f)(g) = \sum_i f(g\gamma_i), \ \ f \in S_{\tau,\psi,\xi}(U, A).$$

We define Hecke algebra $\mathbb{T}^\Sigma_{\tau,\psi,\xi}(U, A) \subseteq \mathrm{End}(S_{\tau,\psi,\xi}(U, A))$ to be the A-subalgebra generated by all $T_v, v \notin \Sigma$. This is a finite commutative A-algebra. We will write $\mathbb{T}^\Sigma_{\psi,\xi}(U, A)$ for the Hecke algebra if τ is the trivial action on A. Suppose that $A = \mathcal{O}/\pi^n$, U is sufficiently small and $\psi\,|_{U\cap(\mathbb{A}_F^\infty)^\times}$ is trivial modulo π^n. Then we can check that the Hecke algebra $\mathbb{T}^\Sigma_{\psi,\xi}(U, \mathcal{O}/\pi^n)$ is independent of Σ, and we may drop Σ in this situation. See [Pan22, 3.3.3].

Definition 4.1.1. (Big Hecke algebra) Let U^p be a tame level and $\psi : (\mathbb{A}_F^\infty)^\times/(U^p \cap (\mathbb{A}_F^\infty)^\times)F_+^\times \to \mathcal{O}^\times$ be a continuous character. Then we define the *Hecke algebra*

$$\mathbb{T}_{\psi,\xi}(U^p) = \varprojlim_{(n,U_p)\in\mathcal{I}} \mathbb{T}_{\psi,\xi}(U^p U_p, \mathcal{O}/\pi^n),$$

where $\mathcal{I}$ is the set of pairs (n, U_p) with $U_p \subseteq K_p$ and n a positive integer such that $\psi\,|_{U_p\cap\mathcal{O}_{F,p}^\times} \equiv 1 \mod \pi^n$.

4.2. **Deformation rings and Hecke algebras.** In this subsection, we collect some results about deformation rings and Hecke algebras, most of which are from [Pan22].

Assume F is an abelian totally real number field of even degree over $\mathbb{Q}$ in which p splits completely. As in Section 3.2, we assume that for all $v \in \Sigma \setminus \Sigma_p$, we have $p \mid \mathrm{Nm}(v) - 1$. Let $\xi_v : k(v)^\times \to \mathcal{O}^\times$ be characters of p-power order for $v \in \Sigma \setminus \Sigma_p$, and we can view them as characters of I_v by local class field theory. Let $\chi : \mathrm{Gal}(F_\Sigma/F) \to \mathcal{O}^\times$ be a continuous totally odd character such that

1) χ is unramified outside Σ_p,
2) $\chi(\mathrm{Frob}_v) \equiv 1 \mod \pi$ for $v \in \Sigma \setminus \Sigma_p$.

Let D be a quaternion algebra over F ramified exactly at all infinite places, and we fix an isomorphism between $(D \otimes_F \mathbb{A}_F^\infty)^\times$ and $\mathrm{GL}_2(\mathbb{A}_F^\infty)$. Let $\psi = \chi\epsilon$ be a character of $(\mathbb{A}_F^\infty)^\times/F_+^\times$

via global class field theory. Let $U^p = \prod_{v\nmid p} U_v$ be a tame level such that $U_v = \mathrm{GL}_2(\mathcal{O}_{F_v})$ if $v \notin \Sigma$ and $U_v = \mathrm{Iw}_v := \{g \equiv \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \bmod \pi_v, g \in \mathrm{GL}_2(\mathcal{O}_{F_v})\}$ otherwise. For any $v \in \Sigma \backslash \Sigma_p$, the map $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \to \xi_v(a/d \bmod \pi_v)$ defines a character of U_v and the product of ξ_v can be viewed as a character ξ of U^p by projecting to $\prod_{v\in\Sigma\backslash\Sigma_p} U_v$. From our setting, we can define a Hecke algebra $\mathbb{T}_\xi := \mathbb{T}_{\psi,\xi}(U^p)$ as in the previous subsection.

From now on, we suppose that $\bar{\rho}_0 = 1 \oplus \bar{\chi}$ is modular, i.e. $T_v - (1 + \chi(\mathrm{Frob}_v)), v \notin \Sigma$ and π generate a maximal ideal $\mathfrak{m}_\xi$ of $\mathbb{T}_\xi$.

Theorem 4.2.1. *Each irreducible component of $(\mathbb{T}_\xi)_{\mathfrak{m}_\xi}$ is of characteristic zero and of dimension at least $1 + 2[F : \mathbb{Q}]$.*

Proof. See [Pan22, Theorem 3.6.1]. □

Remark 4.2.2. In some cases, the big Hecke algebra is equidimensional of dimension $1+2[F : \mathbb{Q}]$. See the proof of [Pan22, Theorem 3.6.1] and [Pan22, Remark 3.6.10].

As the discussion in [Pan22, 3.3.4], there is a natural pseudo-deformation $T_{\mathfrak{m}_\xi} : \mathrm{Gal}(F_\Sigma/F) \to (\mathbb{T}_\xi)_{\mathfrak{m}_\xi}$ with determinant χ sending Frob_v to Hecke operator T_v for $v \notin \Sigma$. By the universal property, we get a natural surjection $R^{\mathrm{ps}}_{\mathrm{aux}} \twoheadrightarrow (\mathbb{T}_\xi)_{\mathfrak{m}_\xi}$. By the local-global compatibility at $v \in \Sigma \setminus \Sigma_p$, we know that such surjection factors through $R^{\mathrm{ps},\{\xi_v\}}$. Hence, we also get a natural surjection $R^{\mathrm{ps},\{\xi_v\}} \twoheadrightarrow (\mathbb{T}_\xi)_{\mathfrak{m}_\xi}$.

For any $\mathfrak{p} \in \mathrm{Spec}(\mathbb{T}_\xi)_{\mathfrak{m}_\xi}$, we can define a two-dimensional semi-simple representation $\rho(\mathfrak{p}) : \mathrm{Gal}(F_\Sigma/F) \to \mathrm{GL}_2(k(\mathfrak{p}))$ with trace $T_{\mathfrak{m}_\xi} \bmod \mathfrak{p}$ in the sense of Definition 2.1.3.

The following definitions are from [Pan22, Section 4.1].

Definition 4.2.3. (1) We say that a prime of $R^{\mathrm{ps},\{\xi_v\}}$ is *pro-modular* if it comes from a prime of $(\mathbb{T}_\xi)_{\mathfrak{m}_\xi}$. We say that a prime of $R^{\mathrm{ps}}_{\mathrm{aux}}$ is *pro-modular* if it comes from a pro-modular prime of some $R^{\mathrm{ps},\{\xi_v\}}$.

(2) Let $\mathfrak{q}$ be a prime of $(\mathbb{T}_\xi)_{\mathfrak{m}_\xi}$ and A be the normal closure of $(\mathbb{T}_\xi)_{\mathfrak{m}_\xi}/\mathfrak{q}$ in $k(\mathfrak{q})$. We say that $\mathfrak{q}$ is a *nice* prime if $\mathfrak{q}$ contains p and $\dim(\mathbb{T}_\xi)_{\mathfrak{m}_\xi}/\mathfrak{q} = 1$ and there exists a two-dimensional representation $\rho(\mathfrak{q})^o : \mathrm{Gal}(F_\Sigma/F) \to \mathrm{GL}_2(A)$ satisfying:

1) $\rho(\mathfrak{q})^o \otimes k(\mathfrak{q}) \cong \rho(\mathfrak{q})$ is irreducible.

2) The mod $\mathfrak{m}_A$ reduction $\bar{\rho}_b$ of $\rho(\mathfrak{q})^o$ is a non-split extension and has the form $\bar{\rho}_b(g) = \begin{pmatrix} * & * \\ 0 & * \end{pmatrix}$, $g \in \mathrm{Gal}(F_\Sigma/F)$. Here $\mathfrak{m}_A$ is the maximal ideal of A.

3) (dihedral condition) If $\rho(\mathfrak{q})$ is dihedral, namely isomorphic to $\mathrm{Ind}_{G_L}^{G_F}\theta$ for some quadratic extension L of F and continuous character $\theta : G_L \to k(\mathfrak{q})^\times$, then $L \cap F(\zeta_p) = F$, where ζ_p is a primitive p-th root of unity.

4) $\rho(\mathfrak{q})^o \mid_{G_{F_v}} = \bar{\rho}_b \mid_{G_{F_v}}$ for any $v \in \Sigma \setminus \Sigma_p$.

(3) We say that a prime of $R^{\mathrm{ps},\{\xi_v\}}$ is *nice* if it comes from a nice prime of $(\mathbb{T}_\xi)_{\mathfrak{m}_\xi}$.

The following proposition gives some sufficient conditions for the third condition of (2) in Definition 4.2.3.

Proposition 4.2.4. Let $\mathfrak{q}$ be a prime ideal of $R^{\mathrm{ps},\{\xi_v\}}$ containing p such that $R^{\mathrm{ps},\{\xi_v\}}/\mathfrak{q}$ is one-dimensional. Suppose $\rho(\mathfrak{q})$ is irreducible. Then the third condition of (2) in Definition 4.2.3 holds for $\mathfrak{q}$ if one of the following conditions holds:

1) $\bar{\chi}$ is not quadratic.

2) $\bar{\chi} \mid_{G_{F_v}}$ is trivial for some $v \mid p$.

3) There exists a place $v \mid p$ such that $\bar{\chi} \mid_{G_{F_v}}$ is not trivial and $\rho(\mathfrak{q})^o \mid_{G_{F_v}} \cong \begin{pmatrix} \chi_{v,1} & * \\ 0 & \chi_{v,2} \end{pmatrix}$ is reducible. Moreover, $\chi_{v,1}$ is of infinite order.

Proof. See [Pan22, Lemma 4.1.6]. □

The following theorem is a key evidence to prove our pro-modularity argument.

Theorem 4.2.5. $(R_{\mathfrak{q}} = \mathbb{T}_{\mathfrak{q}})$ *Let $\mathfrak{q}$ be a nice prime of $(\mathbb{T}_\xi)_{\mathfrak{m}_\xi}$ and $\mathfrak{q}^{\mathrm{ps}} = \mathfrak{q} \cap R^{\mathrm{ps},\{\xi_v\}}$. Then the natural surjective map $(R^{\mathrm{ps},\{\xi_v\}})_{\mathfrak{q}^{\mathrm{ps}}} \twoheadrightarrow (\mathbb{T}_\xi)_{\mathfrak{q}}$ has nilpotent kernel.*

Proof. See [Pan22, Section 4]. □

Remark 4.2.6. We briefly summarize some main steps of the proof of Theorem 4.2.5. Here we follow most notations as in [Pan22, Section 4], and one can find more details there.

Let $\mathfrak{q}$ be a nice prime of $(\mathbb{T}_\xi)_{\mathfrak{m}_\xi}$. Let B be the topological closure of the $\mathbb{F}$-algebra generated by all the entries of $\rho(\mathfrak{q})^o(\mathrm{Gal}(F_\Sigma/F))$. By Chebotarev's density Theorem, we may find a finite set of primes T' disjoint with Σ such that the entries of $\rho(\mathfrak{q})^o(\mathrm{Frob}_v), v \in T'$, topologically generate B. Let $P = T' \cup \Sigma$.

Let $r = \dim_{k(\mathfrak{q})} H^1(G_{F,P}, \mathrm{ad}^0\,\rho(\mathfrak{q})(1))$ (the cardinality of Taylor-Wiles primes). Let $R_\infty^{\{\xi_v\}} = R_{\mathrm{loc}}^{\{\xi_v\}}[[x_1, ..., x_g]]$, where $R_{\mathrm{loc}}^{\{\xi_v\}}$ is the completed tensor product of local deformation rings determined by primes in P (see [Pan22, Definition 4.2.2]) and $g = r + |P| - [F : \mathbb{Q}] - 1$. Let $S'_\infty = \mathcal{O}[[y_1, ..., y_{4|P|-1}, s'_1, ..., s'_r]]$, and $S_\infty = \mathcal{O}[[y_1, ..., y_{4|P|-1}, s_1, ..., s_r]]$ with an ideal $\mathfrak{a}_1 = (y_1, ..., y_{4|P|-1}, s_1, ..., s_r)$ such that S_∞ is a finite free S'_∞-algebra.

Let $\mathrm{m}_\infty^{\{\xi_v\}}$ and $\mathrm{m}_0^{\{\xi_v\}}$ be the *patched Hecke modules* (see [Pan22, Section 4.7]). Then $\mathrm{m}_\infty^{\{\xi_v\}}$ is a flat S_∞-module and $\mathrm{m}_\infty^{\{\xi_v\}}/\mathfrak{a}_1\mathrm{m}_\infty^{\{\xi_v\}} \cong (\mathrm{m}_0^{\{\xi_v\}})^{\oplus 2^r}$. By Pan's local-global compatibility result

[Pan22, Corollary 3.5.8], $\mathrm{m}_0^{\{\xi_v\}}$ is a finitely generated faithful $\widehat{(\mathbb{T}_\xi)_{\mathfrak{q}}}$-module. By Theorem 4.2.1, we have $\dim_{\widehat{(\mathbb{T}_\xi)_{\mathfrak{q}}}}(\mathrm{m}_0^{\{\xi_v\}}) = \dim \widehat{(\mathbb{T}_\xi)_{\mathfrak{q}}} \geq 2[F:\mathbb{Q}]$ since $\mathfrak{q}$ is of dimension 1.

Let $(R^{\{\xi_v\}})'$ be $(\widehat{R_\infty^{\{\xi_v\}}}_{\mathfrak{q}_\infty^{\{\xi_v\}}} \otimes_{S'_\infty} S_\infty$, where $\mathfrak{q}_\infty^{\{\xi_v\}}$ is the prime of $R_\infty^{\{\xi_v\}}$ corresponding to the nice prime $\mathfrak{q}$. We know that $\widehat{(\mathbb{T}_\xi)_{\mathfrak{q}}}$ is a natural quotient of $(R^{\{\xi_v\}})'$ by mapping S_∞ to $S_\infty/\mathfrak{a}_1 = \mathcal{O}$. Note that $y_1, ..., y_{4|P|-1}, s_1, ..., s_r \in \mathfrak{a}_1$ form a regular sequence of $\mathrm{m}_\infty^{\{\xi_v\}}$. Then we can conclude that

$$\dim_{(R^{\{\xi_v\}})'}(\mathrm{m}_\infty^{\{\xi_v\}}) \geq 4|P| - 1 + r + \dim \widehat{(\mathbb{T}_\xi)_{\mathfrak{q}}} \geq 4|P| - 1 + r + 2[F:\mathbb{Q}].$$

Using Taylor's Ihara avoidance trick as in [Tay08], one can show that $\widehat{(R_\infty^{\{\xi_v\}})_{\mathfrak{q}_\infty^{\{\xi_v\}}}}$ is equidimensional of dimension $4|P| - 1 + r + 2[F:\mathbb{Q}]$ (see [Pan22, Section 4.2, Lemma 4.8.2]). As $(R^{\{\xi_v\}})'$ is finite free over $\widehat{(R_\infty^{\{\xi_v\}})_{\mathfrak{q}_\infty^{\{\xi_v\}}}}$, we have

$$\dim_{(R^{\{\xi_v\}})'}(\mathrm{m}_\infty^{\{\xi_v\}}) = \dim_{\widehat{(R_\infty^{\{\xi_v\}})_{\mathfrak{q}_\infty^{\{\xi_v\}}}}}(\mathrm{m}_\infty^{\{\xi_v\}}) = 4|P| - 1 + r + 2[F:\mathbb{Q}] = \dim \widehat{(R_\infty^{\{\xi_v\}})_{\mathfrak{q}_\infty^{\{\xi_v\}}}}.$$

If the characters in $\{\xi_v\}$ are all non-trivial, by [Pan22, Proposition 4.2.3], $\widehat{(R_\infty^{\{\xi_v\}})_{\mathfrak{q}_\infty^{\{\xi_v\}}}}$ is irreducible. From the previous equations, we know that $\mathrm{m}_\infty^{\{\xi_v\}}$ has full support on $\widehat{(R_\infty^{\{\xi_v\}})_{\mathfrak{q}_\infty^{\{\xi_v\}}}}$. By [Pan22, Lemma 4.8.4], Theorem 4.2.5 holds in this case. Similar to the proof of [Tay08, Theorem 4.1], one can deduce the general case using Taylor's trick.

From Theorem 4.2.1, [Eis13, Corollary 12.5], and discussions above, we can also conclude that for a nice prime $\mathfrak{q}$ of $(\mathbb{T}_\xi)_{\mathfrak{m}_\xi}$, we have $\dim \widehat{(\mathbb{T}_\xi)_{\mathfrak{q}}} = \dim(\mathbb{T}_\xi)_{\mathfrak{q}} = 2[F:\mathbb{Q}]$. In other words, if an irreducible component of the big Hecke algebra contains a nice prime, then it is of dimension $1 + 2[F:\mathbb{Q}]$.

Remark 4.2.7. From the definitions and the "$R = \mathbb{T}$" theorem above, we can conclude the following two facts.

1) If a minimal prime $\mathfrak{P} \in \operatorname{Spec} R^{\mathrm{ps},\{\xi_v\}}$ is pro-modular, then any prime contained in the irreducible component defined by $\mathfrak{P}$ is pro-modular. We also say that an irreducible component of $\operatorname{Spec} R^{\mathrm{ps},\{\xi_v\}}$ is pro-modular if its generic point is pro-modular.

2) If an irreducible component of $\operatorname{Spec} R^{\mathrm{ps},\{\xi_v\}}$ contains a nice prime, then it is pro-modular.

4.3. **The ordinary case.** In this subsection, we recall some results on the deformation rings and Hecke algebras in the ordinary case.

Assume that F is an abelian totally real number field over $\mathbb{Q}$ in which p is unramified. Let Σ be a finite set of primes of F containing all places above p. Let $\chi : \mathrm{Gal}(F_\Sigma/F) \to \mathcal{O}^\times$ be a continuous totally odd character such that

1) $\bar{\chi}$ can be extended to a character of $G_{\mathbb{Q}}$.
2) $\bar{\chi}\mid_{G_{F_v}}$ is not trivial for any $v \mid p$.
3) $\chi\mid_{G_{F_v}}$ is de Rham for any $v \mid p$.

Let $(R^{\mathrm{ps,ord}}, T^{\mathrm{ord}})$ be the universal pseudo-deformation ring of $\mathrm{tr}(\bar{\rho}_0)$ of auxiliary type such that $T^{\mathrm{ord}}\mid_{G_{F_v}}$ is reducible for any $v \mid p$, i.e. $y(\sigma,\tau)=0$ for any $\sigma,\tau \in G_{F_v}$, $v \mid p$.

Since $\bar{\chi}\mid_{G_{F_v}}$ is not trivial, we have $T^{\mathrm{ord}}\mid_{G_{F_v}} = \psi_{v,1}^{\mathrm{ord}} + \psi_{v,2}^{\mathrm{ord}}$ for some characters $\psi_{v,1}^{\mathrm{ord}}, \psi_{v,2}^{\mathrm{ord}} : G_{F_v} \to (R^{\mathrm{ps,ord}})^{\times}$ which are liftings of $\mathbf{1}, \bar{\chi}\mid_{G_{F_v}}$ respectively, and actually, $\psi_{v,1}^{\mathrm{ord}} = a\mid_{G_{F_v}}$ and $\psi_{v,2}^{\mathrm{ord}} = d\mid_{G_{F_v}}$ in the sense of Definition 2.1.1. Hence, $\psi_{v,1}^{\mathrm{ord}}\mid_{I_v}$ induces a homomorphism $\mathcal{O}[[\mathcal{O}_{F_v}^{\times}(p)]] \to R^{\mathrm{ps,ord}}$ for any $v \mid p$ via local class field theory, where $\mathcal{O}_{F_v}^{\times}(p)$ is the p-adic completion of $\mathcal{O}_{F_v}^{\times}$.

Definition 4.3.1. [Pan22, Definition 5.3.1] Let $T : \mathrm{Gal}(F_\Sigma/F) \to R$ be a two-dimensional pseudo-representation over some ring R such that for some place v, $T\mid_{G_{F_v}} = \psi_1 + \psi_2$ is a sum of two characters. We say T is ψ_1-*ordinary* if for any $\sigma,\tau \in G_{F_v}$, $\eta \in \mathrm{Gal}(F_\Sigma/F)$,

$$T(\sigma\tau\eta) - \psi_1(\sigma)T(\tau\eta) - \psi_2(\tau)T(\sigma\eta) + \psi_1(\sigma)\psi_2(\tau)T(\eta) = 0.$$

Proposition 4.3.2. Let $\rho : \mathrm{Gal}(F_\Sigma/F) \to \mathrm{GL}_2(R)$ be a two-dimensional irreducible representation over some ring R with trace T such that $T\mid_{G_{F_v}} = \psi_1 + \psi_2$, a sum of two characters. Suppose $\rho\mid_{G_{F_v}}$ has the form $\begin{pmatrix} \psi_1 & * \\ 0 & \psi_2 \end{pmatrix}$. Then T is ψ_1-ordinary. Conversely, if R is a field and T is ψ_1-ordinary, then after possibly enlarging R, we have $\rho\mid_{G_{F_v}} \cong \begin{pmatrix} \psi_1 & * \\ 0 & \psi_2 \end{pmatrix}$.

Proof. See [Pan22, Lemma 5.3.2]. □

Define the *Iwasawa algebra*

$$\Lambda_F := \widehat{\bigotimes}_{v|p} \mathcal{O}[[\mathcal{O}_{F_v}^{\times}(p)]],$$

and we get a natural map $\Lambda_F \to R^{\mathrm{ps,ord}}$ induced by the characters $\psi_{v,1}^{\mathrm{ord}}$. The following result is the ordinary Fontaine-Mazur conjecture proved in [SW99] and [Pan22, Section 5], which is used to produce pro-modular primes in our setting.

Proposition 4.3.3. (1) $R^{\mathrm{ps,ord}}$ is a finite Λ_F-algebra.

(2) For any maximal ideal $\mathfrak{p}$ of $R^{\mathrm{ps,ord}}[\frac{1}{p}]$, we denote the associated semi-simple representation $\mathrm{Gal}(F_\Sigma/F) \to \mathrm{GL}_2(k(\mathfrak{p}))$ by $\rho(\mathfrak{p})$. Assume that

i) $\rho(\mathfrak{p})$ is irreducible.

ii) For any $v \mid p$, $\rho(\mathfrak{p}) \mid_{G_{F_v}} \cong \begin{pmatrix} \chi_{v,1} & * \\ 0 & \chi_{v,2} \end{pmatrix}$ such that $\chi_{v,1}$ is de Rham and has strictly less Hodge-Tate number than $\chi_{v,2}$ for any embedding $F_v \hookrightarrow \overline{\mathbb{Q}}_p$.

Then $\rho(\mathfrak{p})$ comes from a twist of Hilbert modular form.

Proof. See [Pan22, Theorem 5.1.1]. □

Lemma 4.3.4. *Let C^{ord} be an irreducible component of $R^{\mathrm{ps,ord}}$ of dimension at least $1+[F:\mathbb{Q}]$. Let $C^{\mathrm{ord,aut}}$ be the set of regular de Rham primes in C^{ord}. Then $C^{\mathrm{ord,aut}}$ is dense in C^{ord}. Here a prime $\mathfrak{q}$ is called regular de Rham prime if*

i) $p \notin \mathfrak{q}$ and $R^{\mathrm{ps,ord}}/\mathfrak{q}$ is one-dimensional.

ii) The semi-simple representation $\rho(\mathfrak{q}) : \mathrm{Gal}(F_\Sigma/F) \to \mathrm{GL}_2(k(\mathfrak{q}))$ (in the sense of Definition 2.1.1) is irreducible.

iii) For any $v \mid p$, $\rho(\mathfrak{q}) \mid_{G_{F_v}} \cong \begin{pmatrix} \chi_{v,1} & * \\ 0 & \chi_{v,2} \end{pmatrix}$ such that $\chi_{v,1}$ is de Rham and has strictly less Hodge-Tate number than $\chi_{v,2}$ for any embedding $F_v \hookrightarrow \overline{\mathbb{Q}}_p$.

Proof. The proof is the same as [Pan22, Corollary 7.2.3]. We sketch it here.

From our assumption and Proposition 4.3.3, we know that the morphism $C^{\mathrm{ord}} \to \operatorname{Spec} \Lambda_F$.

For $v \mid p$, let $\psi_{v,1}^{\mathrm{ord}}, \psi_{v,2}^{\mathrm{ord}} : G_{F_v} \to (R^{\mathrm{ps,ord}})^\times$ be the liftings of $1, \bar{\chi} \mid_{G_{F_v}}$ respectively. By Proposition 4.3.2, there exists $n_v \in \{1,2\}$ such that for any $\mathfrak{q} \in C_1^{\mathrm{ord}}$, we get $\rho(\mathfrak{q}) \mid_{G_{F_v}} \cong \begin{pmatrix} \psi_{v,n_v}^{\mathrm{ord}} \bmod \mathfrak{q} & * \\ 0 & \psi_{v,3-n_v}^{\mathrm{ord}} \bmod \mathfrak{q} \end{pmatrix}$.

Note that the map $\Lambda_F \to R^{\mathrm{ps,ord}}$ is induced by the characters $\psi_{v,1}^{\mathrm{ord}}$. Hence, we know that the image of $C^{\mathrm{ord,aut}}$ is dense in $\operatorname{Spec} \Lambda_F$, and hence dense in C^{ord}. □

4.4. **The pro-modularity of universal deformation rings.** In this subsection, we will prove our pro-modularity statements.

Keep the notations and assumptions on the finite set Σ and the character χ in the previous two subsections. In this subsection, we further assume that $\bar{\chi} \mid_{G_{F_v}} \neq \omega_p^{\pm 1}$ for any $v \mid p$, and $[F:\mathbb{Q}] \geq 7|\Sigma \setminus \Sigma_p| + 2$.

Definition 4.4.1. We say that a prime of $R_x^{\mathrm{aux},\{\xi_v\}}$ is pro-modular if its inverse image of the natural map $R^{\mathrm{ps},\{\xi_v\}} \to R_x^{\mathrm{aux},\{\xi_v\}}$ is pro-modular. We say that a prime of R_x^{aux} is pro-modular if it is the inverse image of some pro-modular prime of some $R_x^{\mathrm{aux},\{\xi_v\}}$ via the natural surjection $R_x^{\mathrm{aux}} \twoheadrightarrow R_x^{\mathrm{aux},\{\xi_v\}}$.

Theorem 4.4.2. *For any irreducible component of $R_{\mathrm{aux}}^{\mathrm{ps}}$ of dimension at least $1+2[F:\mathbb{Q}]$, it is pro-modular.*

Proof. Let C be an irreducible component of $R^{\mathrm{ps}}_{\mathrm{aux}}$ of dimension at least $1+2[F:\mathbb{Q}]$. Consider the corresponding generic point $\mathfrak{P}$ of $R^{\mathrm{ps}}_{\mathrm{aux}}$, and we suppose that it belongs to $\{\xi_v\}$. We claim that $\mathfrak{P}$ is pro-modular.

For convenience for readers, we briefly introduce our strategy here. By Remark 4.2.7, we just need to find a nice prime containing $\mathfrak{P}$. In other words, we need to find a pro-modular prime which satisfies the conditions we list in (2) of Definition 4.2.3. Hence, our proof can be divided into two parts. In the first part, following [Pan22, Section 7], we use Pan's results given in the previous section to find enough pro-modular primes from ordinary points. (By the way, there is also a similar step in the proof of Deo's big $R=\mathbb{T}$ theorem. See [Deo23, Proposition 3.4].) In the second part, we use Corollary 2.1.7 to find a nice prime, which is a main innovation of this paper.

Write $C^{\mathrm{ord}}=\operatorname{Spec} R^{\mathrm{ps,ord}}\cap C$.

Lemma 4.4.3. $\dim C^{\mathrm{ord}}\geq [F:\mathbb{Q}]+1$.

Proof. For any $v\mid p$, let R^{ps}_v be the universal pseudo-deformation ring of the pseudo-representation of G_{F_v} lifting $1+\bar{\chi}\mid_{G_{F_v}}$ of auxiliary type, i.e. of determinant $\chi\mid_{G_{F_v}}$. Let $R^{\mathrm{ps,ord}}_v$ be the quotient of R^{ps}_v parametrizing all reducible liftings. As $\bar{\chi}\neq 1,\omega_p^{\pm 1}$, we know that the kernel of $R^{\mathrm{ps}}_v\twoheadrightarrow R^{\mathrm{ps,ord}}_v$ is a principal ideal by [Paš13, Corollary B.20]. Write $R^{\mathrm{ps}}_p=\widehat{\bigotimes}_{\mathcal{O}}R^{\mathrm{ps}}_v$ and $R^{\mathrm{ps,ord}}_p=\widehat{\bigotimes}_{\mathcal{O}}R^{\mathrm{ps,ord}}_v$. Then by universal property, we get $R^{\mathrm{ps,ord}}=R^{\mathrm{ps}}_{\mathrm{aux}}\otimes_{R^{\mathrm{ps}}_p}R^{\mathrm{ps,ord}}_p$. Thus, $\dim C^{\mathrm{ord}}\geq \dim C-|\Sigma_p|\geq 1+2[F:\mathbb{Q}]-[F:\mathbb{Q}]=[F:\mathbb{Q}]+1$. □

Now we can choose an irreducible component $C^{\mathrm{ord}}_1\subset C^{\mathrm{ord}}$ of $\operatorname{Spec} R^{\mathrm{ps,ord}}$ of dimension $1+[F:\mathbb{Q}]$, and we let $\mathfrak{Q}$ be the generic point of C^{ord}_1. For any $v\in\Sigma\setminus\Sigma_p$, we fix a lifting σ_v of the generator of the $\mathbb{Z}_p$-quotient of I_v and a lifting Frob_v of the Frobenius element. Write $S=\{\sigma_v,\mathrm{Frob}_v:\ v\in\Sigma\setminus\Sigma_p\}$. Clearly, $|S|=2|\Sigma\setminus\Sigma_p|$.

Let I be an ideal of $R^{\mathrm{ps}}_{\mathrm{aux}}/(\mathfrak{Q},\pi)$ generated by

$$\{a(\sigma_v)-1,a(\mathrm{Frob}_v)-1,d(\mathrm{Frob}_v)-1,\ v\ \in\Sigma\setminus\Sigma_p\}.$$

We consider a minimal prime $\mathfrak{u}$ of I. By Krull's principal ideal theorem, $(R^{\mathrm{ps}}_{\mathrm{aux}}/(\mathfrak{Q},\pi))/\mathfrak{u}$ is of dimension at least $[F:\mathbb{Q}]-3|\Sigma\setminus\Sigma_p|$. For simplicity, we write such a domain as R_0.

To find a nice prime, we consider the following steps.

Step I. We start from the CNL domain R_0.

By Corollary 2.1.7, we can find a partition of $S = S_1 \amalg S_2$, a positive integer $n > 1$ prime to p and a CNL domain R_0' satisfying the following conditions.

1) R_0' is a quotient of R_0.

2) For any $\theta \in S_1$, we have $y(\theta, \alpha) = 0$ for any $\alpha \in \mathrm{Gal}(F_\Sigma/F)$ in R_0'. For any $\theta', \theta'' \in S_2$, we have $y(\theta', \alpha)^n = y(\theta'', \alpha)^n$ for any $\alpha \in \mathrm{Gal}(F_\Sigma/F)$ in R_0'.

3) For any $\alpha \in \mathrm{Gal}(F_\Sigma/F), \theta \in S_2$, either $y(\theta, \beta) = 0$ for any $\beta \in \mathrm{Gal}(F_\Sigma/F)$ or $u(\alpha, \theta)$ is well-defined and integral over R_0'.

4) We have $\dim R_0' \geq \dim R_0 - |S|$. If S_2 is not empty, then further $\dim R_0' \geq \dim R_0 - |S| + 1$.

Step II. We start from the CNL domain R_0'.

Similar to *Step* I, we can find a partition of $S = S_1' \amalg S_2'$, a positive integer $n' > 1$ prime to p and a CNL domain R_0'' satisfying the following conditions.

1') R_0'' is a quotient of R_0'.

2') For any $\theta \in S_1'$, we have $y(\alpha, \theta) = 0$ for any $\alpha \in \mathrm{Gal}(F_\Sigma/F)$ in R_0''. For any $\theta', \theta'' \in S_2'$, we have $y(\alpha, \theta')^{n'} = y(\alpha, \theta'')^{n'}$ for any $\alpha \in \mathrm{Gal}(F_\Sigma/F)$ in R_0''.

3') For any $\alpha \in \mathrm{Gal}(F_\Sigma/F), \theta \in S_2'$, either $y(\beta, \theta) = 0$ for any $\beta \in \mathrm{Gal}(F_\Sigma/F)$ or $u'(\alpha, \theta)$ is well-defined and integral over R_0''.

4') We have $\dim R_0'' \geq \dim R_0' - |S|$. If S_2' is not empty, then further $\dim R_0'' \geq \dim R_0' - |S| + 1$.

Step III. We consider the domain R_0''.

If both S_2 and S_2' are not empty, then we fix an element $\theta_i \in S_2$ and an element $\theta_j \in S_2'$. Consider a prime $\mathfrak{p}_{ij}$ of R_0'' of the ideal $(y(\theta_i, \theta_j))$ and let $R'' = R_0''/\mathfrak{p}_{ij}$. By Proposition 2.1.2, we know that either $y(\theta_i, \alpha) = 0$ or $y(\beta, \theta_j) = 0$ for all $\alpha, \beta \in \mathrm{Gal}(F_\Sigma/F)$. By Krull's principal ideal theorem, we have $\dim R'' \geq \dim R_0'' - 1$.

If not, we take $R'' = R_0''$.

In conclusion, one of the following three assertions holds for R''.

a) $y(\theta, \alpha) = y(\alpha, \theta) = 0$ for all $\theta \in S$ and $\alpha \in \mathrm{Gal}(F_\Sigma/F)$.

b) There exists an element $\theta_i \in S$ such that either $y(\theta, \alpha) = 0$ or $y(\theta, \alpha)^n = y(\theta_i, \alpha)^n$ for all $\theta \in S$ and $\alpha \in \mathrm{Gal}(F_\Sigma/F)$, and $y(\alpha, \theta) = 0$ for all $\theta \in S$ and $\alpha \in \mathrm{Gal}(F_\Sigma/F)$.

c) There exists an element $\theta_j \in S$ such that either $y(\alpha, \theta) = 0$ or $y(\alpha, \theta)^{n'} = y(\alpha, \theta_j)^{n'}$ for all $\theta \in S$ and $\alpha \in \mathrm{Gal}(F_\Sigma/F)$, and $y(\theta, \alpha) = 0$ for all $\theta \in S$ and $\alpha \in \mathrm{Gal}(F_\Sigma/F)$.

In particular, we have $\dim R'' \geq [F : \mathbb{Q}] - 7|\Sigma \setminus \Sigma_p| \geq 2$.

Lemma 4.4.4. *There exists a nice prime containing* $\mathfrak{P}$.

Proof. By Proposition 4.3.3, for any prime $\mathfrak{q}_1 \in C_1^{\mathrm{ord,aut}}$, the subset of C_1^{ord} consisting of all regular de Rham primes as defined in Lemma 4.3.4, we know that $\rho(\mathfrak{q}_1)$ comes from a twist of Hilbert modular form. By the density result in Lemma 4.3.4, the image of any prime of C_1^{ord} in $\operatorname{Spec} R^{\mathrm{ps},\{\xi_v\}}$ is pro-modular.

By Corollary 3.2.3, we can choose an irreducible one-dimensional prime $\mathfrak{r}'$ of R''. We have shown that $\mathfrak{r}'$ is pro-modular. Now we claim that $\mathfrak{r}'$ is actually a nice prime in the sense of Definition 4.2.3.

Let A be the normal closure of $R''/\mathfrak{r}'$. Then we obtain a continuous irreducible representation $\rho(\mathfrak{r}') : \mathrm{Gal}(F_\Sigma/F) \to \mathrm{GL}_2(A)$ associated to the pseudo-deformation in the following form

$$\rho(\mathfrak{r}')(\sigma) = \begin{pmatrix} a(\sigma) & \frac{y(\sigma,\tau_{\mathfrak{r}'})}{y(\sigma_{\mathfrak{r}'},\tau_{\mathfrak{r}'})} \\ y(\sigma_{\mathfrak{r}'},\sigma) & d(\sigma) \end{pmatrix},\ y(\sigma_{\mathfrak{r}'},\tau_{\mathfrak{r}'}) \neq 0.$$

Now we verify the conditions in Definition 4.2.3 for all the three cases.

In case a), we can take $\rho(\mathfrak{r}')^o = \rho(\mathfrak{r}')$. From our construction, the first three conditions are clear (using 3) of Proposition 4.2.4). By Proposition 3.2.1, we know that if $a(\sigma_v) = 1$, then we have $d(\sigma_v) = 1$ for any $v \in \Sigma \setminus \Sigma_p$ as $T(\sigma_v) = \xi_v + \xi_v^{-1} \equiv 2 \mod \pi$. Thus, we can find that for all $v \in \Sigma \setminus \Sigma_p$, $\rho(\mathfrak{r}')^o \mid_{G_{F_v}} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. Then $\mathfrak{r}'$ is a nice prime.

In case b), if $y(\theta_i, \alpha) = 0$ for all $\alpha \in \mathrm{Gal}(F_\Sigma/F)$, then it is the same as case a). If not, by Proposition 2.1.2, we know that $y(\theta_i, \tau_{\mathfrak{r}'}) \neq 0$, otherwise $y(\sigma_{\mathfrak{r}'}, \tau_{\mathfrak{r}'}) = 0$. By 3) in *Step* I, $u(\sigma_{\mathfrak{r}'}, \theta_i)$ is well-defined and integral over $R''/\mathfrak{r}'$, and hence a unit in A. Then we can take $\rho(\mathfrak{r}')^o = P\rho(\mathfrak{r}')P^{-1}$, where

$$P = \begin{pmatrix} u(\sigma_{\mathfrak{r}'}, \theta_i) & 0 \\ 0 & 1 \end{pmatrix} \in \mathrm{GL}_2(A).$$

Similar to the case a), we only need to check the fourth condition. From our construction, we know that either $u(\theta, \sigma_{\mathfrak{r}'}) = 0$ or $u(\theta, \theta_i)^n = 1$ for any $\theta \in S$. As $(n, p) = 1$, by Hensel's lemma, $u(\theta, \theta_i)$ is an element of $\mathbb{F}' \hookrightarrow \mathbb{F}'[[T]] = A$, where $\mathbb{F}'$ is the residue field of A. Thus, the fourth condition holds.

In case c), we consider the continuous irreducible Galois representation $\rho(\mathfrak{r}')' : \mathrm{Gal}(F_\Sigma/F) \to \mathrm{GL}_2(A)$ in the following form

$$\rho(\mathfrak{r}')'(\sigma) = \begin{pmatrix} a(\sigma) & y(\sigma,\tau_{\mathfrak{r}'}) \\ \frac{y(\sigma_{\mathfrak{r}'},\sigma)}{y(\sigma_{\mathfrak{r}'},\tau_{\mathfrak{r}'})} & d(\sigma) \end{pmatrix},\ y(\sigma_{\mathfrak{r}'},\tau_{\mathfrak{r}'}) \neq 0.$$

If $y(\alpha, \theta_j) = 0$ for all $\alpha \in \mathrm{Gal}(F_\Sigma/F)$, then we can take

$$\rho(\mathfrak{r}')^o = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \rho(\mathfrak{r}')' \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \subset \mathrm{GL}_2(A).$$

Similar to case a), we can vefiry that $\mathfrak{r}'$ is a nice prime. If not, we can take

$$\rho(\mathfrak{r}')^o = \begin{pmatrix} 0 & u'(\tau_{\mathfrak{r}'}, \theta_j) \\ 1 & 0 \end{pmatrix} \rho(\mathfrak{r}')' \begin{pmatrix} 0 & 1 \\ u'(\theta_j, \tau_{\mathfrak{r}'}) & 0 \end{pmatrix} \subset \mathrm{GL}_2(A).$$

Similar to case b), we can also vefiry that $\mathfrak{r}'$ is a nice prime. □

From the previous lemma and Theorem 4.2.5, we know that C is pro-modular.

□

Remark 4.4.5. 1) By Proposition 3.2.7, for a one-dimensional prime $\mathfrak{p}$ of $R^{\mathrm{ps}}_{\mathrm{aux}}$, if it corresponds to an irreducible pseudo-representation, then at least one of the irreducible components of $R^{\mathrm{ps}}_{\mathrm{aux}}$ containing $\mathfrak{p}$ is of dimension at least $1 + 2[F : \mathbb{Q}]$. Hence $\mathfrak{p}$ is pro-modular.

2) In the proof of Theorem 4.4.2, we can also find that the irreducible component C is of dimension $1 + 2[F : \mathbb{Q}]$. Note that the Iwasawa algebra Λ_F is of dimension $1 + [F : \mathbb{Q}]$. By the finiteness result (Proposition 4.3.3), we have $\dim C^{\mathrm{ord}} \leq [F : \mathbb{Q}] + 1$. Combining Lemma 4.4.3, we have $\dim C^{\mathrm{ord}} = [F : \mathbb{Q}] + 1$, and hence $\dim C = 2[F : \mathbb{Q}] + 1$.

As a consequence of our pro-modularity argument, we can get a conditional $R = \mathbb{T}$ theorem.

Theorem 4.4.6. *If we further assume*

$$[F : \mathbb{Q}] \geq \max\{1 + \frac{1}{2}\dim_{\mathbb{F}}(H^1(F_\Sigma/F, \mathbb{F}(\bar{\chi}^{-1}))), 7|\Sigma \setminus \Sigma_p| + 2\},$$

then for any non-zero element $x \in H^1(F_\Sigma/F, \mathbb{F}(\bar{\chi}^{-1}))$, any prime ideal of R^{aux}_x is pro-modular. Moreover, R^{aux}_x is a local complete intersection ring of Krull dimension $1 + 2[F : \mathbb{Q}]$.

Proof. Let $\mathfrak{P}$ be a minimal prime of R^{aux}_x. Let $\mathfrak{P}'$ be the inverse image of $\mathfrak{P}$ via the natural map $f : R^{\mathrm{ps}}_{\mathrm{aux}} \to R^{\mathrm{aux}}_x$. From our assumptions, Proposition 3.1.3, Proposition 3.1.4 and Proposition 3.2.5, we know that $\mathfrak{P}'$ is a minimal prime of dimension at least $1 + 2[F : \mathbb{Q}]$. By Theorem 4.4.2, we know that $\mathfrak{P}$ is pro-modular.

By Remark 4.4.5, we know that $\mathfrak{P}'$ is of dimension $1 + 2[F : \mathbb{Q}]$. Consequently, R^{aux}_x is of dimension at most $1 + 2[F : \mathbb{Q}]$. Combining Proposition 3.1.2 and Proposition 3.1.3, we obtain that R^{aux}_x is a local complete intersection ring of Krull dimension $1 + 2[F : \mathbb{Q}]$. This proves our result.

□

4.5. **A special case.** In this subsection, we deal with the case $\bar{\chi}\ |_{G_{F_v}} = \omega_p$ for any $v|p$. In this case, the deduction is more complicated because Lemma 4.4.3 no longer holds. We follow the strategy of the proof in [Pan22, Section 7.4].

Keep assumptions on as in Section 4.2 and Section 4.3 the finite set Σ and the character χ in the previous two subsections. We further assume $[F:\mathbb{Q}] \geq 7|\Sigma \setminus \Sigma_p| + 4$.

Proposition 4.5.1. Let $\mathfrak{r}$ be a one-dimensional irreducible pro-modular prime of $R^{\mathrm{ps}}_{\mathrm{aux}}$. Then every irreducible component of $R^{\mathrm{ps}}_{\mathrm{aux}}$ containing $\mathfrak{r}$ contains a nice prime, and hence is pro-modular.

Proof. Let Z be the set of all irreducible components of $(R^{\mathrm{ps}}_{\mathrm{aux}})_{\mathfrak{r}}$. Write $Z = Z_1 \amalg Z_2$, where Z_1 is the subset of Z consisting of all pro-modular irreducible components. We first show that if Z_1 is not empty, then Z_2 is empty.

If not, by Proposition 3.2.7, there exist an irreducible component $Y \in Z_2$ and a pro-modular prime $\mathfrak{r}' \in Y$ of dimension at least $2[F:\mathbb{Q}] - 1$.

Arguing as the proof of Theorem 4.4.2, we may find a prime $\mathfrak{r}'' \subset \mathfrak{r}'$ satisfying the following conditions:

- $\varpi \in \mathfrak{r}''$.
- The dimension of $\mathfrak{r}''$ is at least $2[F:\mathbb{Q}] - 7|\Sigma \setminus \Sigma_p| - 2 \geq [F:\mathbb{Q}] + 2$.
- For any irreducible one-dimensional prime $\mathfrak{t}$ containing $\mathfrak{r}''$, there exists a 2-dimensional Galois representation $\rho(\mathfrak{t}) : G_{F,\Sigma} \to \mathrm{GL}_2(k(\mathfrak{t}))$ satisfying the conditions in (2) of Definition 4.2.3 except the third one.

If $\bar{\chi}$ is not quadratic, then by Lemma 4.2.4, we can find a nice prime containing $\mathfrak{r}''$ (hence containing $\mathfrak{r}'$). Using Theorem 4.2.5, we conclude that Y is pro-modular, which contradicts to our definition of Z_2.

If $\bar{\chi}$ is quadratic, we know that the splitting field $F(\bar{\chi})$ is an abelian CM field. Then by Corollary 3.2.3, we may find a prime containing $\mathfrak{r}'$, whose corresponding Galois representation is irreducible and not dihedral. From our choice of $\mathfrak{r}''$, we know that this is actually a nice prime. Using Theorem 4.2.5 again, we get a contradiction.

As $\mathfrak{r}$ is pro-modular, using Corollary 4.2.1, we know that there exists a pro-modular prime $\mathfrak{r}_1$ contained in $\mathfrak{r}$ of dimension at least $1 + 2[F:\mathbb{Q}]$. Arguing as above, we can find a nice prime containing $\mathfrak{r}_1$. This implies that at least one of the irreducible component containing $\mathfrak{r}$ is pro-modular. Hence, Z_1 is not empty. □

From now on, we assume $\bar{\chi}\ |_{G_{F_v}} = \omega_p$ for any $v|p$. Suppose that C is an irreducible component of $R^{\mathrm{ps}}_{\mathrm{aux}}$ of dimension at least $1 + 2[F:\mathbb{Q}]$. In this case, by [Pan22, Lemma 7.4.4],

the kernel of the natural surjection $R_v^{\mathrm{ps}} \twoheadrightarrow R_v^{\mathrm{ps,ord}}$ can be generated by two elements. Then by Krull's principal ideal theorem, we have $\dim C^{\mathrm{ord}} \geq 1$.

Choose a prime $\mathfrak{q} \in C^{\mathrm{ord}}$ such that $\dim R_{\mathrm{aux}}^{\mathrm{ps}}/\mathfrak{q} = 1$. Enlarging $\mathcal{O}$ if necessary, we may assume the normalization of $R_{\mathrm{aux}}^{\mathrm{ps}}/\mathfrak{q}$ is either $\mathcal{O}$ or isomorphic to $\mathbb{F}[[T]]$. There are three possibilities for the associated semi-simple representation $\rho(\mathfrak{q})$:

(1) $\rho(\mathfrak{q})$ is irreducible.
(2) $\rho(\mathfrak{q}) \cong \psi_1 \oplus \psi_2$ is reducible and ψ_1/ψ_2 is not of the form $\epsilon^{\pm 1}\theta$, where θ is a finite order character of G_F. We call this case *generic reducible*.
(3) $\rho(\mathfrak{q}) \cong \psi_1 \oplus \psi_2$ and $\psi_1/\psi_2 = \varepsilon\theta$, for some finite order character θ of G_F. We call this case *non-generic reducible*.

Theorem 4.5.2. *The irreducible component C is pro-modular of dimension $1 + 2[F : \mathbb{Q}]$.*

Proof. As the discussions above, we divide the proof into three cases.

1. The irreducible case

Using the same proof of [Pan22, Lemma 7.4.7], we find that the localization $(R^{\mathrm{ps,ord}})_{\mathfrak{q}}$ is of dimension at least $[F : \mathbb{Q}]$. In other words, there exists an irreducible component of $R^{\mathrm{ps,ord}}$ containing $\mathfrak{q}$ of dimension at least $1 + [F : \mathbb{Q}]$. Arguing as in the proof of Theorem 4.4.2, we know that $\mathfrak{q}$ is pro-modular. By Proposition 4.5.1 and Remark 4.2.6, we know that C is pro-modular of dimension $1 + 2[F : \mathbb{Q}]$.

2. The generic reducible case

By the same proof as in [Pan22, Section 7.4.4], we have $\dim C^{\mathrm{ord}} \geq 1 + [F : \mathbb{Q}]$ (see [Pan22, Corollary 7.4.9]). Thus, as in the irreducible case, we also have C is pro-modular of dimension $1 + 2[F : \mathbb{Q}]$.

3. The non-generic reducible case

We only sketch the proof.

For any non-zero class $B \in \mathrm{Ext}^1_{E[G_{F,\Sigma}]}(\psi_1, \psi_2) \cong H^1(G_{F,\Sigma}, E(\psi_2/\psi_1))$, it corresponds to a non-split extension $\rho_B : G_{F,\Sigma} \to \mathrm{GL}_2(E)$ of ψ_1 by ψ_2. We may assume ρ_B is of the form

$$\begin{pmatrix} \psi_2 & b_B \\ 0 & \psi_1 \end{pmatrix}.$$

We denote the universal deformation ring of ρ_B with fixed determinant χ by R_B.

We define Z_B as the set of irreducible components of $R_{\mathrm{aux}}^{\mathrm{ps}}$ whose generic points lie in the image of $\mathrm{Spec}\, R_B \to \mathrm{Spec}\, R_{\mathrm{aux}}^{\mathrm{ps}}$.

First, by the arguments in [Pan22, Step (1), page 1153], there exists a non-zero element B_0 in $H^1(G_{F,\Sigma}, E(\psi_2/\psi_1))$ such that we can find a pro-modular irreducible component in

Z_{B_0}. Using [Pan22, Corollary 7.4.13] and Proposition 4.5.1, we know that any irreducible component in Z_{B_0} is pro-modular.

Second, using the same proof in [Pan22, Step (2), page 1154 & 1155], we can find an element B_1 in $H^1(G_{F,\Sigma}, E(\psi_2/\psi_1))$ such that $C \in Z_{B_1}$ and there exist irreducible components $C_{B_0} \in Z_{B_0}, C_{B_1} \in Z_{B_1}$ containing a prime of dimension $\mathfrak{s}$ at least $2 + [F : \mathbb{Q}]$. As C_{B_0} is pro-modular, we know that $\mathfrak{s}$ is pro-modular. By Corollary 3.2.3, we can find a one-dimensional irreducible pro-modular prime in $C_{B_0} \cap C_{B_1}$. Combining Proposition 4.5.1, we know that C_{B_1} is pro-modular. Using [Pan22, Corollary 7.4.13] and Proposition 4.5.1 again, we know that any irreducible component in Z_{B_1} is pro-modular. In particular, C is pro-modular of dimension $1 + 2[F : \mathbb{Q}]$. □

Remark 4.5.3. 1) If $\bar{\chi}$ is quadratic, then we cannot ignore the dihedral condition in the defintion of a nice prime. By Corollary 3.2.2, the upper bound of the dihedral locus is $1+[F : \mathbb{Q}]$, so we may not find a nice prime from the pro-modular prime $\mathfrak{s}$ as its dimension may not be large enough. However, we can find one-dimensional irreducible pro-modular prime from $\mathfrak{s}$ and then use Proposition 4.5.1, which is the advantage of our method (particularly Corollary 2.1.7) in this paper.

2) We can also prove the same results as Theorem 4.4.6 using the same proof.

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